CTA-2034 in Simulated Rooms: What the Listener Receives, Direction by Direction

The spinorama's weights against 300 domestic listening rooms, and a two-way loudspeaker with Linkwitz-Riley and brickwall crossovers

Technical report — simulation code and data:
https://github.com/Arnwald/cta2034-room-study

Abstract

The curves of ANSI/CTA-2034-A (Listening Window, Early Reflections, Sound Power and the Predicted In-Room Response, PIR) are weighted means of the power a loudspeaker radiates in 70 directions on two orbits. The energy reaching a listener in a room is also a weighted mean of the power radiated in every direction, with weights set by the paths' lengths and absorptions. We compute those weights in 300 simulated domestic rooms (specular images up to the third order and a diffuse field fed direction by direction; reverberation time 0.40 s at 1 kHz, median) and compare them with the standard's, using a two-way bookshelf loudspeaker whose far field is known over the whole sphere, with a plain dome tweeter or a tweeter in a waveguide matched to the woofer at the crossover, and LR2, LR4, LR8 or brickwall FIR crossovers. Three hypotheses were tested: that the standard under-weighs the vertical directions, that the Listening Window hides the lobing of low-slope crossovers, and that a weighting faithful to the received energy would show a brickwall crossover removing most of the off-axis colouration. The floor and ceiling bring 41 % of the first reflections' energy (50 % where the side walls are treated), the share the Early Reflections give them, and their reflections leave the loudspeaker at angles the standard brackets; the Listening Window's asymmetry matches where listeners sit. The weights themselves are off: the first reflections carry 11 % of the steady-state energy, against 44 % in the PIR, and the rooms are predicted to within their own spread by about $0.11\,\mathrm{LW} + 0.89\,\mathrm{SP}$, where the PIR is 1.7 to 2.1 dB RMS off; the two-orbit sound power shows about half of the power lost to lobing. Averaging over directions, in the standard's curves as in the room, fills the notches the crossover digs into each reflection: the floor and ceiling bounces of an LR4 two-way carry notches of 10 to 40 dB a third of an octave wide, which the averages reduce to a 2 dB dip, and the averages rank LR2 as the smoothest Linkwitz-Riley crossover while its reflections are the most coloured. Read path by path, a brickwall crossover removes about 60 % of the colouration of the floor and ceiling bounces (59 % with the plain dome, 60 % with the waveguide), leaving notches as deep but ten to thirteen times narrower; the in-room curve becomes smoother only when the drivers' directivities match at the crossover. Added with their phases at the listener's ears and smoothed by the auditory filters, the paths change the steady state little; in the early sound at a fixed point the comb of the floor and ceiling bounces outweighs any crossover's colouration, and the Linkwitz-Riley notch, which takes these reflections out near the crossover, removes their comb there.

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1 Introduction

ANSI/CTA-2034-A [1] describes a loudspeaker by 70 far-field measurements: a horizontal and a vertical orbit around a reference axis, every 10°. Its curves average them: the *Listening Window* (LW) over the directions toward a group of listeners, the *Early Reflections* (ER) over the directions of the first reflections in a typical room, the *Sound Power* (SP) over all 70 directions, each weighted by the solid angle it stands for. Their mix, the *Predicted In-Room Response*

$$\mathrm{PIR} = 0.12\,\mathrm{LW} + 0.44\,\mathrm{ER} + 0.44\,\mathrm{SP}, \tag{1}$$

estimates the steady-state response measured at the listening position. The curves and the prediction come from Devantier's work at Harman [3], the reflection directions from a survey of domestic listening rooms; they are central to the work of Toole and Olive on loudspeaker preference [4, 5].

The standard was written for the measurement systems of its time, two orbits on a turntable, and much of its practice grew around passive loudspeakers. Three objections motivated this study:

1. The vertical directions weigh less than the horizontal ones (two of the nine directions of the Listening Window, two of the five bounces of the Early Reflections), while domestic rooms are wider than high, and floors and ceilings are harder to treat than walls.

2. The Listening Window, ±30° horizontally but ±10° vertically, would hide the vertical lobing of the low-slope crossovers of passive loudspeakers.

3. A weighting closer to the energy actually received would reveal that a brickwall crossover, which leaves the drivers overlapping over a narrow band, removes most of the off-axis colouration of a Linkwitz-Riley crossover [6].

Each of the standard's curves is a weighted mean of the power responses in the 70 directions. A room does the same: in steady state, and once the paths are added in power (which a spatial average over a few tenths of a metre achieves above the room's transition frequency), the energy at the listener is a sum over paths of the power the loudspeaker radiates in each path's direction times the path's weight. The comparison is thus between two sets of weights over directions. We compute the rooms' in 300 simulated domestic listening rooms (section 4), with a loudspeaker whose far field is known over the whole sphere (section 3), and compare them with the standard's (section 5): where the energy leaves the loudspeaker, how much of it each kind of path carries, how well the PIR predicts the rooms, and what an updated weighting gives. The loudspeaker then comes with four crossovers, to test the third objection (section 6).

The curves follow CTA-2034-A as it is commonly implemented, the Early Reflections being the mean of its five bounces. CTA-2034-B was published in July 2024 [2]; its changes were not checked.

2 The standard's curves as weights over directions

The two orbits pass through the reference axis. In the loudspeaker's frame (x to the right as seen from the listener, y up, z forward), the horizontal orbit at angle h (positive toward $+x$) and the vertical orbit at angle v (positive upward) are the directions

$$\mathbf{u}_h = (\sin h,\, 0,\, \cos h), \qquad \mathbf{u}_v = (0,\, \sin v,\, \cos v), \tag{2}$$

every 10°; 0° and 180° are shared, which leaves 70 directions. With $P_d(f)$ the far-field power in direction d, every curve is

$$C(f) = \sum_{d=1}^{70} w_d^C \, P_d(f), \qquad \sum_d w_d^C = 1 : \tag{3}$$

- LW: 1/9 on $h = 0, \pm10, \pm20, \pm30°$ and $v = \pm10°$;
- ER: the mean of five bounces, each the mean of its directions: floor ($v = -20, -30, -40°$), ceiling ($v = 40, 50, 60°$), front wall ($h = 0, \pm10, \pm20, \pm30°$), side walls ($h = \pm40 \ldots \pm 80°$), rear wall ($h = \pm90°, 180°$);
- SP: each direction at angle θ from the axis weighted by the solid angle between the "spherical squares" of half-widths $\theta - 5°$ and $\theta + 5°$, the squares bounded by the great circles at $\pm\alpha$ horizontally and vertically, $S(\alpha) = 4C(\alpha) - 2\pi$ with C the square's corner angle; the 90° band is counted twice.

The PIR's weights are $0.12\, w^{\mathrm{LW}} + 0.44\, w^{\mathrm{ER}} + 0.44\, w^{\mathrm{SP}}$ (figure 4, top). A loudspeaker radiating the same power in every direction has the same value in every curve.

The SP weights make two orbits stand for the sphere: the response at $h = \theta$ or $v = \theta$ stands for a whole square ring, corners included. For a source symmetric about its axis the corners lie farther off axis than the orbit, so a directional source is overestimated slightly: by 0.45 dB for a piston of $ka = 3$ with the obliquity of a baffle, a few hundredths of a decibel at low frequency. For two drivers stacked vertically the horizontal orbit sees no interference at all, but only a thin band of the sphere around it is spared (section 5.4).

3 The simulated loudspeakers

3.1 Layout and radiators

The loudspeaker is an upright two-way bookshelf, 200 × 360 × 280 mm, its two drivers on the vertical middle line of the front baffle: a 6.5-inch cone, a flat rigid piston of radius 64 mm (129 cm^2) centered 30 mm below the baffle's center, and a 1.1-inch dome 140 mm above it, the rigid spherical

cap of the turntable study [7] (radius 14 mm, height 7.5 mm). The crossover is at 2.2 kHz: there the cone's ka is 2.58 and it beams, the dome's is 0.56 and it does not.

A variant puts the dome in a waveguide. A waveguide is modelled phenomenologically as a source with the directivity of a piston whose size stops growing in wavelengths above a given frequency:

$$D(\theta, k) = \frac{2J_1(x)}{x}, \qquad x = \min(ka, \kappa)\sin\theta, \tag{4}$$

with $a = 64$ mm and $\kappa = 2.58$: below 2.2 kHz it radiates like the cone, above it keeps the cone's beamwidth at the crossover (about 115° between the −6 dB points). It shields the baffle's edges accordingly (below).

3.2 Far field with cabinet diffraction

The cabinet's diffraction follows the Distributed Edge Dipole model [8] as implemented in the turntable study [7]: a line of dipoles along the edges of the front baffle, driven by the pressure each driver produces there. Here it is evaluated in the far field, in any direction $\mathbf{u}$, relative to the baffle's center and normalized to 1 m:

$$H_i(\mathbf{u}, f) = D_i(\theta, k)\,\frac{1+\cos\theta}{2}\,e^{\mathrm{j}k\,\mathbf{u}\cdot\mathbf{s}_i} - u_z \sum_e Q_{i,e}(f)\,e^{\mathrm{j}k\,\mathbf{u}\cdot\mathbf{E}_e}, \qquad \cos\theta = u_z, \tag{5}$$

$\mathbf{s}_i$ the driver's center, $\mathbf{E}_e$ the edge elements (2 mm), $Q_{i,e}$ their strengths. The direct term vanishes behind the cabinet, the edge dipoles change sign at the baffle plane. At low frequency the loudspeaker radiates 6 dB below its half-space response, at high frequency it reaches it. For the waveguide the edge strengths are the dome's times $D(90°, k)$. The far field agrees with the near-field model of the turntable study evaluated at 300 m within 0.6 % of its largest value.

3.3 Equalization and crossovers

Each driver i is equalized to an acoustic target T_i on the reference axis (the baffle normal): its filter is $X_i = T_i E_i$, E_i the minimum-phase inverse of its 1/6-octave smoothed on-axis magnitude, which removes the baffle step and the broad part of the edge ripple. The targets are Linkwitz-Riley of order 2, 4 or 8 (squared Butterworth of order 1, 2 or 4, the LR2's high-pass inverted so that both outputs are in phase at the crossover), or the brickwall linear-phase FIR crossover of the four-way article [6]: a windowed sinc of 1501 taps at 93.75 kHz (Kaiser window, $\beta = 8$) and its complement, an impulse minus the low-pass. Its transition is about 320 Hz wide, its stopband 81 dB down. Every variant is flat on axis within 0.25 dB from 300 Hz to 16 kHz. The loudspeaker's power in any direction is then

$$P(\mathbf{u}, f) = \Big|\sum_i X_i(f)\,H_i(\mathbf{u}, f)\Big|^2. \tag{6}$$

At the crossover frequency all four crossovers give each driver half the amplitude, in phase; they differ by the width of the band over which both drivers play (figure 1).

For the rotation test the loudspeaker is also laid on its side, its drivers side by side, the reference axis unchanged: $P_{\text{side}}(\mathbf{u}) = P(R^{\mathsf{T}}\mathbf{u})$, R the rotation by 90° about the axis.

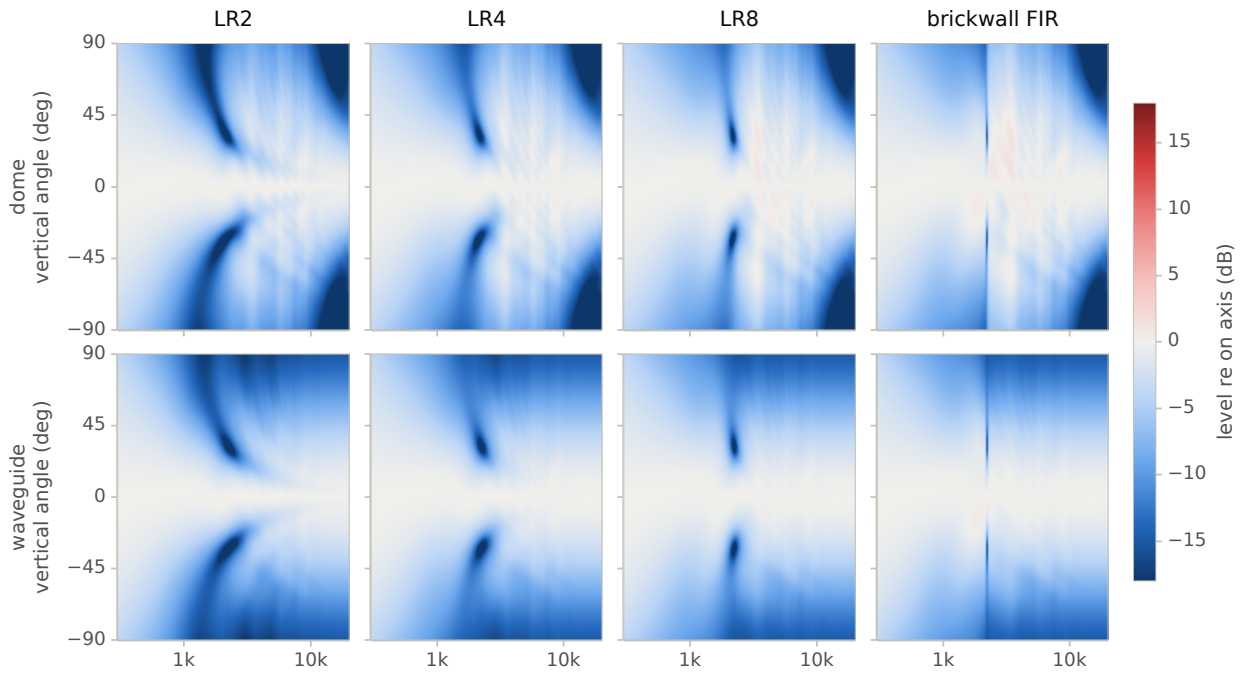


Figure 1: Vertical-plane directivity of the two-way, relative to the on-axis response, with the four crossovers: plain dome (top), dome in a waveguide (bottom).

4 The rooms

4.1 Model

The rooms are shoeboxes. In the room's frame x runs across the room, y from the wall behind the listener to the wall behind the loudspeakers, z up. The sound reaching the listener is split three ways, all in energy.

Specular images up to the third order [9]. On each axis, with walls at 0 and L, the image of the source s is $u = 2nL + (1-2q)s$ (n an integer, $q \in \{0,1\}$), having bounced $|n-q|$ times on the wall at 0 and $|n|$ times on the wall at L. The loudspeaker emits that path in the direction $\mathbf{P}(\mathbf{r}-\mathbf{u})/|\mathbf{r}-\mathbf{u}|$, $\mathbf{P}$ flipping the axes with an odd number of bounces ($q=1$), $\mathbf{r}$ the listener's head. Its weight is

$$W = \frac{e^{-m|\mathbf{r}-\mathbf{u}|}}{|\mathbf{r}-\mathbf{u}|^2} \prod_{\text{bounces}} (1-\alpha)(1-s), \tag{7}$$

α the absorption coefficient, s the scattering coefficient (the share of the reflected energy sent in all directions instead of the specular one), m the energy attenuation per metre of the air [12], and of the furniture beyond the first order.

Diffuse field. A ray launched from the loudspeaker in direction Ω is traced through the box. At each of its first three bounces the scattered share $(1-\alpha)s$ of the energy it carries joins a diffuse field; after the third, everything left does. This defines $g(\Omega) \in [0,1]$, the share of the energy emitted in direction Ω that feeds the diffuse field. In steady state that field gives everywhere

$$p^2_{\text{diff}} = \frac{4}{A} \oint g(\Omega)\, P(\Omega)\, \mathrm{d}\Omega, \qquad A = -S \ln(1-\bar{\alpha}) + 4mV, \tag{8}$$

with the Eyring absorption area of the whole room, air and furniture included [10]. With $g = 1-\alpha$ everywhere this is the room term $4/R$ of Hopkins and Stryker [11]. The integral runs over 3000 directions of a Fibonacci lattice.

Direct sound, the image of order 0.

Absorbing patches (a rug, panels, a ceiling cloud) replace a surface's coefficients where a ray or a first-order reflection lands on them; the higher-order images and the diffuse field use each surface's area-weighted mean. The furniture is an equivalent absorption area A_f spread over the volume, an attenuation $m_f = A_f/4V$ per metre applied to the images of order 2 and more, to the rays and to the diffuse field; the direct sound and the first reflections are assumed unobstructed.

The energy at the listener is then

$$E(f) = \sum_{\text{images}} W_k(f)\, P(\Omega_k, f) + \sum_{\text{lattice}} \frac{4}{A}\,\frac{4\pi}{N}\, g(\Omega_l, f)\, P(\Omega_l, f), \tag{9}$$

a set of weights over directions, like the standard's curves. We give it relative to an omnidirectional source at the same place ($P \equiv 1$), which divides out the room's own colouration (its absorption rising with frequency): $E(f)/E_{\text{omni}}(f)$ is how the room weighs the loudspeaker's directions, the quantity the PIR estimates. A loudspeaker radiating the same everywhere gets 0 dB in every room. Two versions: the *steady state*, every path, and the *early sound*, the direct sound and the six first reflections, what the LW and the ER stand for.

4.2 Checks

In a room with uniform absorption the model agrees with Hopkins–Stryker when every reflection scatters ($s = 1$), within 1 %. When none does ($s = 0$), its total energy is within 3 to 7 % (0.15 to 0.3 dB) of a full image method summed to the 26th or 40th order (25 000 and 88 000 images), for $\bar{\alpha} = 0.25$ and 0.1. The rooms' scattering coefficients (0.05 to 0.5) lie between these two cases.

4.3 The ensemble

Each setup holds the left loudspeaker of a stereo pair and a listener on the room's middle line (the right loudspeaker is its mirror image). The 300 random setups (table 1) draw: width 3.4–5.2 m, length 4.5–7.5 m, height 2.4–2.8 m; loudspeakers on the short wall, their baffles 0.4–1.2 m from it, 1.8–2.8 m apart and at least 0.45 m from the side walls, the reference axis 0.85–1.15 m high; listener 2.0–3.8 m from the baffles' line and at least 0.5 m in front of the wall behind, ears 1.0–1.25 m high; loudspeakers aimed at the listener (half the setups), parallel to the side walls (a quarter) or halfway (a quarter). Walls of painted plaster on masonry or plasterboard on studs, one with a window (2–6 m^2) and curtains (0–4 m^2); ceiling of plaster or plasterboard; floor of parquet or tiles (60 %, half of them with a rug 2–3 × 2.5–3.5 m in front of the loudspeakers) or wall-to-wall carpet; furniture absorbing 6–16 m^2 at 1 kHz, half that at 125 Hz. A quarter of the rooms (72) is treated: panels of 50 mm mineral wool, 1.2 × 1.2 m, at the side walls' first reflection points, and a rug; their ceilings stay bare. The absorption and scattering coefficients are typical textbook values; the reverberation times (0.40 s median at 1 kHz, 0.26–0.68 s for 80 % of the rooms) fall in the range measured in furnished dwellings [13]. The first 150 rooms fit the updated weights, the other 150 test them.

Table 1: The 300 random setups: 10th percentile, median, 90th percentile.

	10 %	median	90 %
width (m)	3.60	4.29	5.03
length (m)	4.95	6.05	7.16
height (m)	2.43	2.59	2.75
volume (m^3)	52.06	66.49	85.52
listening distance (m)	2.14	2.85	3.64
loudspeaker spacing (m)	1.91	2.29	2.67
baffle to wall behind (m)	0.48	0.78	1.12
T_{60} at 500 Hz (s)	0.30	0.44	0.73
T_{60} at 1 kHz (s)	0.26	0.40	0.68
T_{60} at 4 kHz (s)	0.24	0.35	0.58

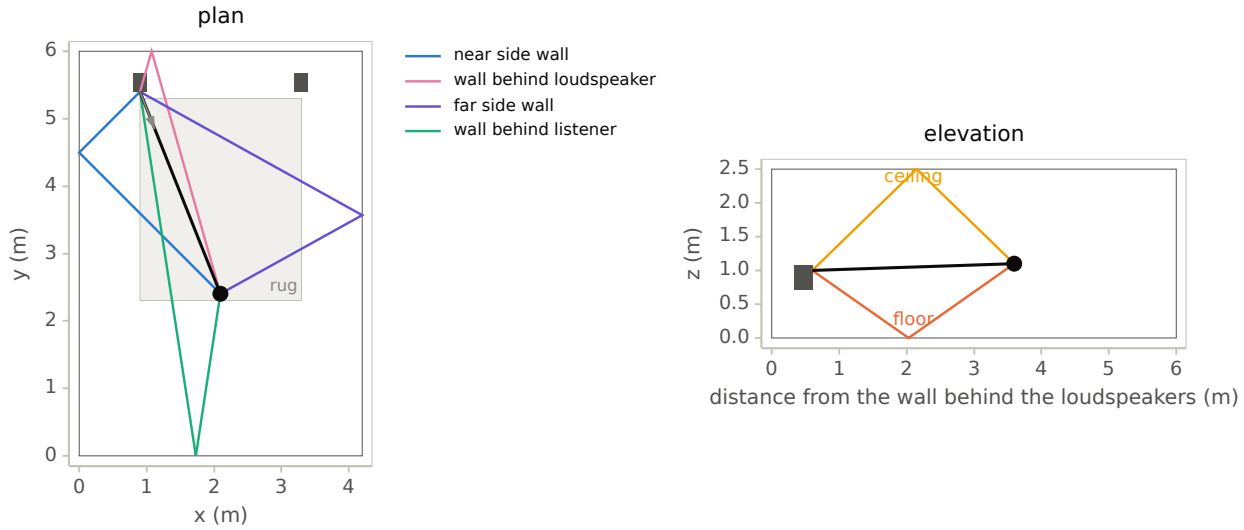


Figure 2: One of the rooms (4.2 × 6.0 × 2.5 m, parquet and rug): the left loudspeaker, the listener, the direct sound (black) and the six first reflections.

5 The standard against the rooms

5.1 Where the energy goes

For an omnidirectional source, the direct sound brings 6.6 % of the steady-state energy at 1 kHz (median), the six first reflections 11 % together, the later paths 82 % (table 2, figure 3). The PIR gives 12 % to the Listening Window, 44 % to the Early Reflections and 44 % to the Sound Power: about twice the direct sound's share and four times the first reflections'.

Among the first reflections the floor and the ceiling bring 41 % of the energy (33–52 %), the side walls 31 %: the share the Early Reflections give them, two bounces of five. In the treated rooms the floor and ceiling rise to 50 % (39 % in the others). The ceiling outweighs the floor: rugs and carpets absorb the floor bounce, and the furniture on the floor scatters it.

5.2 In which directions

The first reflections leave the loudspeaker where the standard looks for them (table 3, figures 4 and 5). The floor bounce leaves at −35° (median), the ceiling bounce at 45°; the standard's floor and ceiling angles bracket them, its −20° and 60° fitting listeners farther or closer than these. Both arrive first, 1.9 and 3.6 ms after the direct sound, with the near side wall (3.5 ms): in rooms wider than high the floor and ceiling are the nearest surfaces. The side walls fall within the standard's

Table 2: Share of the energy at the listener (%), omnidirectional source: median and 10–90 % range over the 300 rooms.

	1 kHz		4 kHz	
	median	range	median	range
direct sound	6.6	3.5–12.2	7.7	4.0–13.7
first order: floor	1.6	1.1–2.3	1.4	1.0–2.0
first order: ceiling	2.7	1.6–5.0	3.0	1.8–5.3
first order: near side wall	2.0	0.0–4.1	1.9	0.2–4.0
first order: far side wall	1.2	0.6–2.2	1.2	0.7–2.1
first order: wall behind listener	0.8	0.4–1.9	0.8	0.4–1.9
first order: wall behind loudspeaker	2.5	1.3–4.5	2.4	1.3–4.4
second and third order	17.7	12.2–21.8	14.9	10.2–17.7
diffuse field	64.9	48.8–76.2	66.6	50.6–77.7

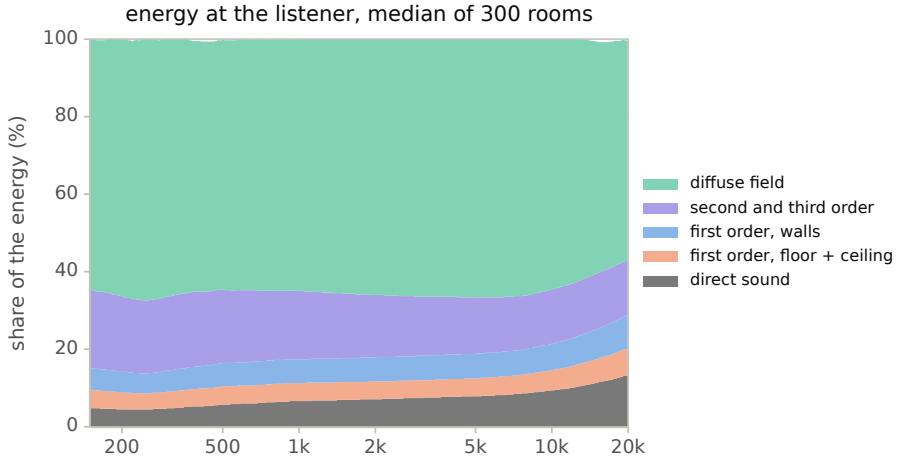


Figure 3: Share of the energy at the listener by kind of path, omnidirectional source, median of the 300 rooms.

±40 to ±80°; the wall behind the listener (the standard's front wall) within its ±30°. The wall behind the loudspeakers (its rear wall) is reached at 137 to 166°, where the standard looks at ±90° and 180°.

The diffuse field does not draw evenly on all directions either (figure 4, bottom): what the loudspeaker sends toward an absorbing floor or a rug feeds it less than what it sends backward or upward. The differences reach about ±30 %, much less than the standard's contrast between the ER directions and the rest.

5.3 Predicting the in-room response

With weights summing to one fitted on the 150 training rooms, all sixteen loudspeaker variants together, the steady-state response is best predicted by

$$0.11\,\mathrm{LW} + 0.01\,\mathrm{ER} + 0.89\,\mathrm{SP}_{\mathrm{sphere}}, \tag{10}$$

with the sound power integrated over the sphere, or by $0.08\,\mathrm{LW} + 0.02\,\mathrm{ER} + 0.90\,\mathrm{SP}$ with the two orbits' (table 4, figure 6). Fitted to subsets (treated rooms, reverberation times below 0.33 or above 0.5 s, listeners closer than 2.6 or farther than 3.2 m), the LW weight stays between 0.07 and 0.16 and the ER weight between −0.05 and 0.04. Fitted on the plain dome alone, the weights predict the waveguide as well as its own fit. The Early Reflections do not help: their directions are those of a minor share of the energy, which the LW and SP already describe. On the test rooms these weights

Table 3: Directions in which the first reflections leave the loudspeaker (degrees, median and 10–90 % range over the 300 rooms), the standard's angles (V vertical, H horizontal orbit) and the reflections' delay after the direct sound (median).

reflection	angle	median	range	CTA-2034	delay (ms)
floor	elevation	−35	−41 to −29	−20, −30, −40 (V)	1.9
ceiling	elevation	45	38 to 52	40, 50, 60 (V)	3.6
near side wall	azimuth	−59	−82 to −46	±40 . . . ± 80 (H)	3.5
far side wall	azimuth	46	38 to 64	±40 . . . ± 80 (H)	9.0
wall behind listener	azimuth	−5	−18 to 10	0, ±10, ±20, ±30 (H)	13.4
wall behind loudspeaker	azimuth	151	137 to 166	±90, 180 (H)	4.4

Table 4: Predicting the steady-state response of the 150 test rooms, 300 Hz–16 kHz: median RMS error over the rooms, and largest mean error (bias) over frequency, range over the sixteen variants (dB).

prediction	RMS error	largest bias
CTA-2034 PIR, 0.12/0.44/0.44, two orbits	1.70–2.11	2.53–2.91
same weights, sound power of the sphere	1.65–1.99	2.21–2.62
fitted, two orbits, 0.08/0.02/0.90	0.44–0.72	0.63–0.98
fitted, sphere, 0.11/0.01/0.89	0.45–0.74	0.41–0.57
mean of the 150 training rooms (room-to-room floor)	0.43–0.69	–

are as close to each room as the mean of the training rooms is, the room-to-room floor. The PIR is 1.7 to 2.1 dB RMS off, mostly by a bias: it sits 0.8 dB above the rooms at 300 Hz and up to 2.6 dB above at the crossover, because it leans on the Listening Window, where a forward-directional loudspeaker is loudest (figure 7).

For the early sound alone the rooms weigh about 0.40 LW + 0.53 ER + 0.07 SP: the direct sound brings 36 % of it, where the early part of the PIR (0.12 LW + 0.44 ER, normalized) gives the Listening Window 21 %.

5.4 The sound power from two orbits

Around the crossover both drivers play, and the lobing between them costs sound power. The upright loudspeaker's horizontal orbit sees no interference, the drivers being at the same distance from it; the vertical orbit sees it. The two orbits get the same weight, where only a thin band of the sphere around the horizontal plane is spared. At 2 kHz the two-orbit sound power is 1.8 dB too high with LR4 (1.8 with LR2, 1.7 with LR8; figure 8): of the 2.6 dB by which the LR4's sound power falls below the brickwall's there, the two orbits show 1.2. The same happens on its side, the orbits trading roles. Above the crossover the two orbits overestimate the directional dome by 0.2 to 0.6 dB, the effect of the spherical squares (section 2).

5.5 Upright or on its side

Laid on its side, the loudspeaker lobes horizontally instead of vertically. The Listening Window sees the difference (figure 9): about 1 dB down at 2 kHz upright, 2.3 to 2.6 dB on its side; the PIR barely moves. The rooms' steady state does not move either: it is mostly sound power, which does not depend on the orientation. The early sound does, though not in its median. Upright, no room leaves the listener more than 1.1 dB below the median at the crossover; on its side, a quarter of the rooms leave the listener more than 3 dB below it, one in ten 5.7 dB: setups where the loudspeakers are not aimed at the listener, who sits 23 to 33° off axis, near the lobing's first null (34° at 2.2 kHz). Seated listeners hear the direct sound within 5° of the axis vertically (90 % of the rooms), and up to 24° horizontally. The Listening Window's ±30° horizontally and ±10° vertically follow that

asymmetry. The vertical lobing reaches the listener through the floor and ceiling bounces and the reverberant field, which the ER and the SP carry, averaged (section 6).

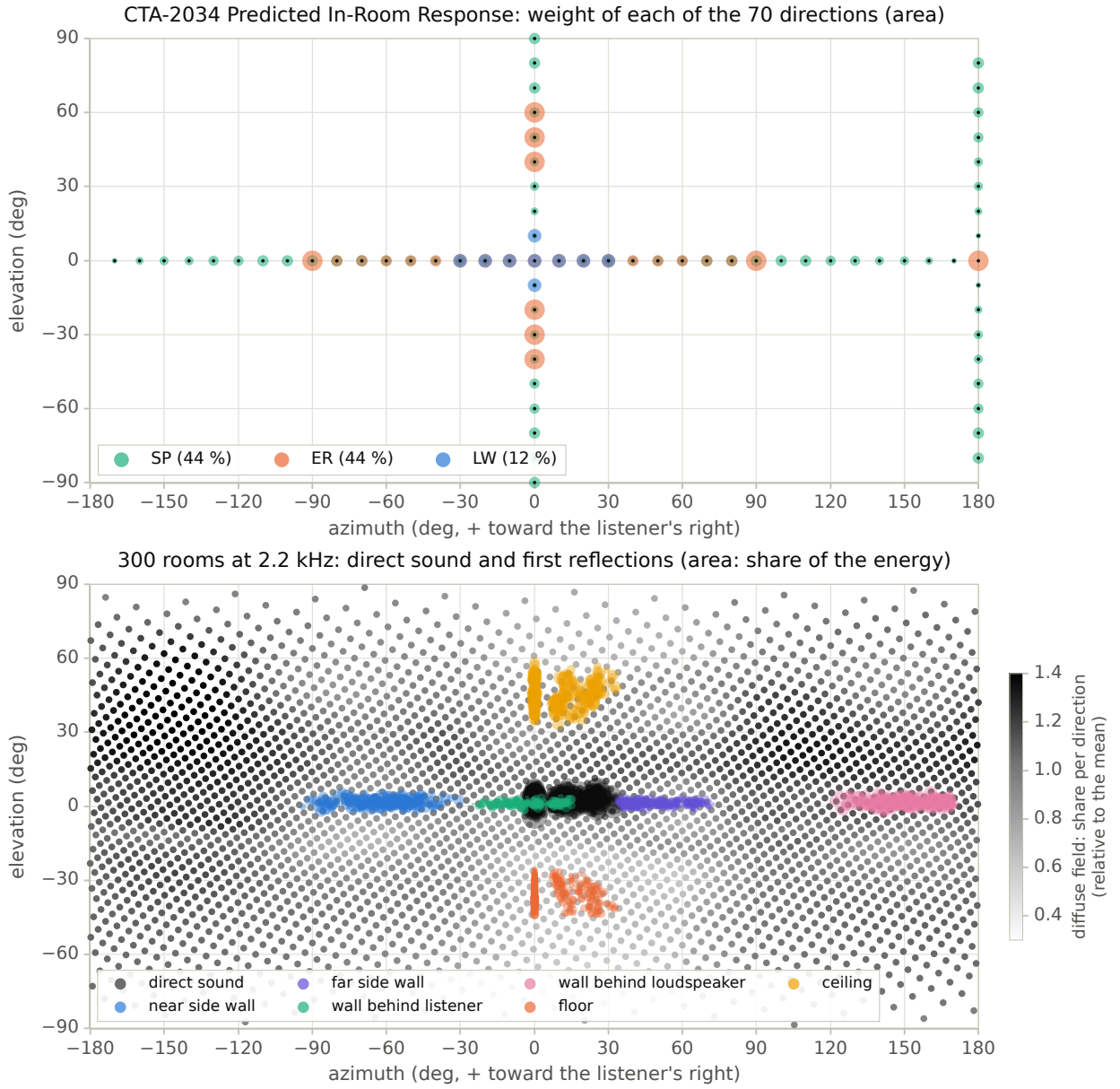


Figure 4: Top: the weight the PIR gives each of the 70 directions (area of the circles), split into its LW, ER and SP parts; beyond 90° the vertical orbit is drawn at azimuth 180°. Bottom: the 300 rooms at 2.2 kHz, the direct sound and the first reflections (area: share of the energy at the listener), over the diffuse field's share per direction g, relative to its mean (grey).

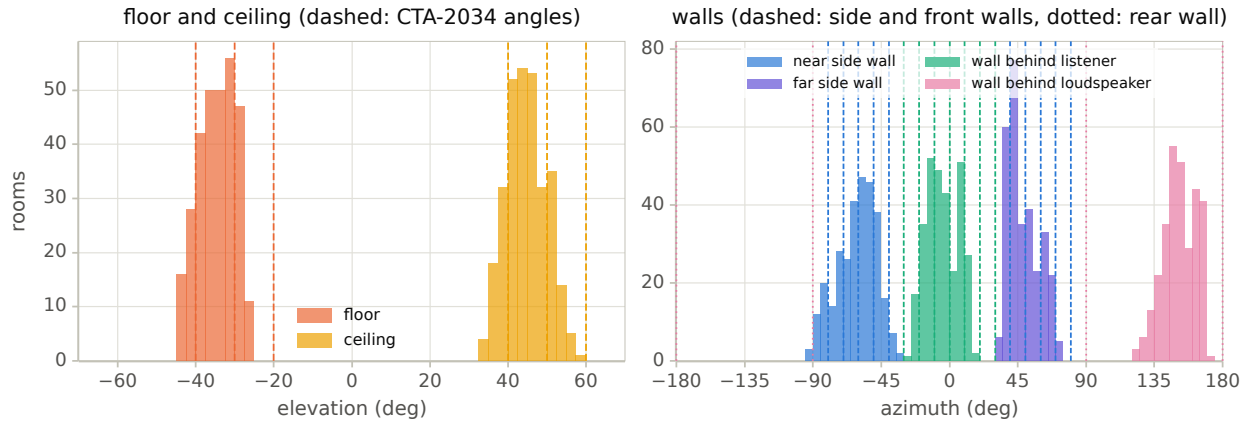


Figure 5: Directions of the first reflections in the 300 rooms, and the standard's angles.

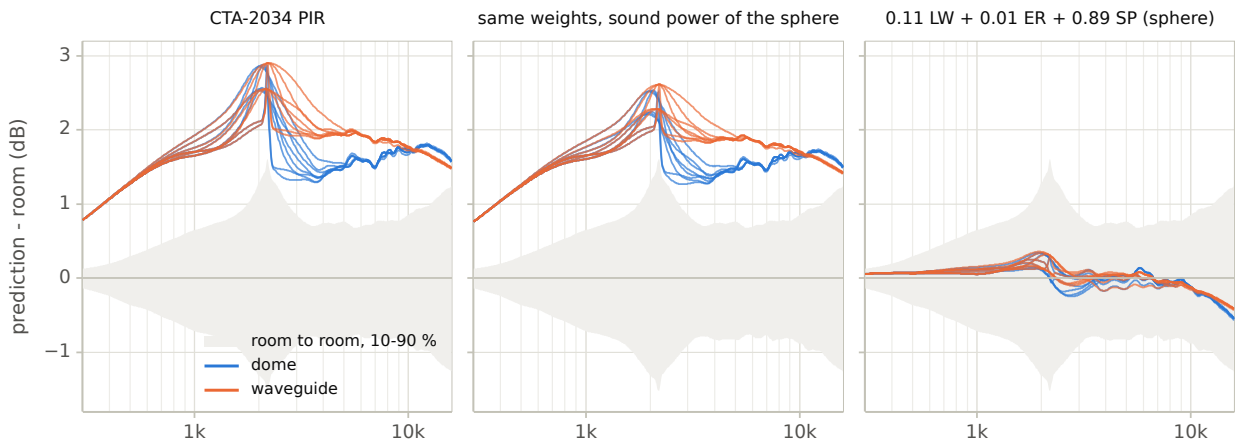


Figure 6: Mean error of three predictions on the 150 test rooms, every variant (thin lines), and the rooms' own spread around their mean (grey).

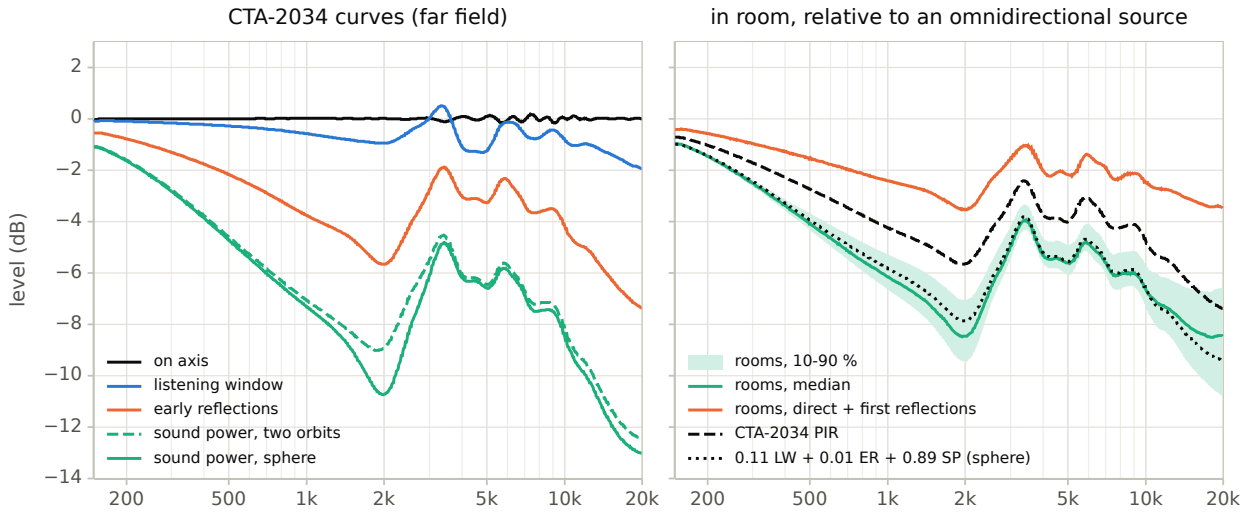


Figure 7: The upright two-way, plain dome, LR4. Left: its CTA-2034 curves, the sound power from two orbits and over the sphere. Right: in the rooms, relative to an omnidirectional source, steady state (median and 10–90 % range) and early sound (median), with the PIR and the fitted weights.

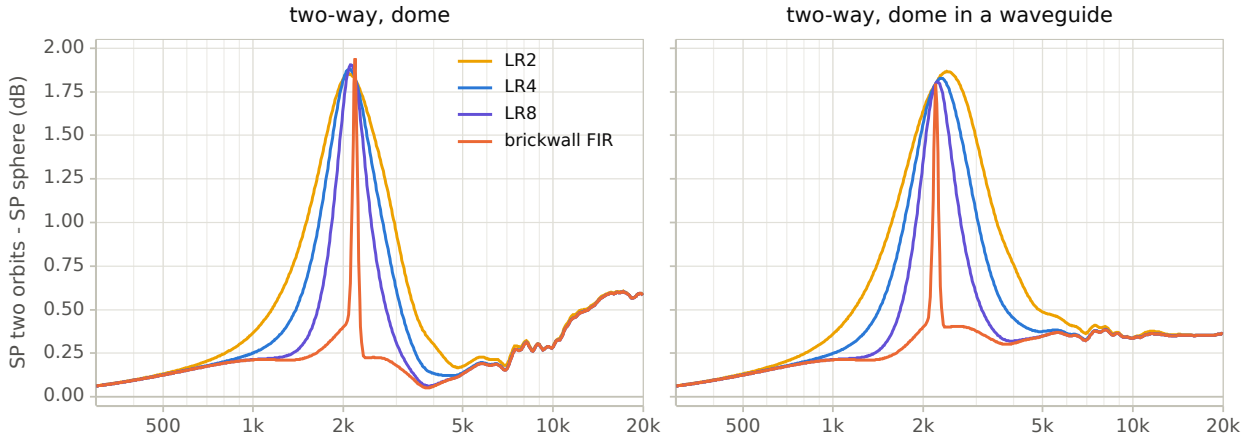


Figure 8: Sound power from the two orbits minus the sound power over the sphere, upright two-way.

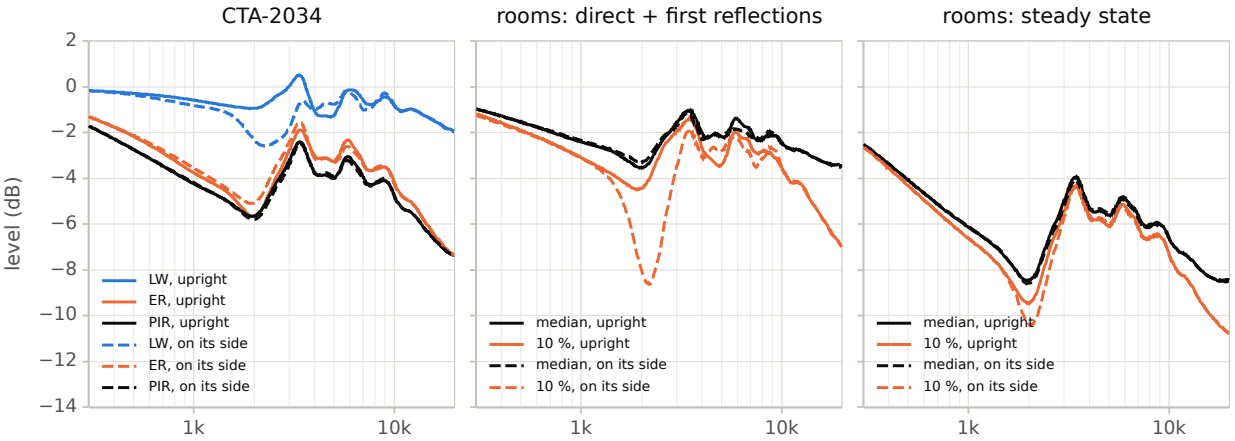


Figure 9: The two-way with LR4, upright and on its side. Left: its LW, ER and PIR. Middle and right: the rooms, early sound and steady state, median and 10th percentile.

6 Linkwitz-Riley against brickwall FIR

6.1 Averages and paths

A crossover's defects lie within about an octave of its frequency, and in each direction at a frequency of its own: two drivers d apart cancel where $d \sin\theta$ is half a wavelength, at 34° for 2.2 kHz and 140 mm, at wider angles for lower frequencies. Adding directions in power, as the in-room curve, the PIR and every curve of the standard do, adds notches that fall at different frequencies, and fills them. Two measures follow.

Averaged curve. The deviation of a curve (dB against log frequency) from the straight line that best fits it from 600 Hz to 8 kHz, a loudspeaker whose directivity index rises evenly giving a straight line; its RMS within one octave of the crossover (1.1 to 4.4 kHz), and the deepest dip there.

Colouration of the paths. The same deviation for the response in each direction, before any averaging, then weighted by the energy that path brings to the listener: the room's weight times the loudspeaker's power in that direction, averaged over 600 Hz–8 kHz and over the four crossovers. The weights are thus the same for all crossovers, so that a notch does not lower the weight of the path it colours. Groups: every path (steady state), the first reflections, the floor and ceiling bounces alone; and the 70 directions with the PIR's weights or the updated ones.

6.2 What the floor and ceiling bounces carry

In the living room of figure 2 the floor bounce leaves at −33°, close to the first null, and the ceiling bounce at +42° (figure 10). With the plain dome and LR4, the floor bounce has a notch 41 dB deep, 611 Hz wide at −6 dB from the straight line (745 Hz with LR2); the brickwall's is 21 dB deep and 48 Hz wide. At the ceiling, farther from the null: 10 dB over 387 Hz with LR4, 8 dB over 32 Hz with the brickwall. With the waveguide the notches are as deep, and the brickwall's as narrow. In the same room's averaged response the LR4 leaves a dip of 2 to 3 dB, about an octave wide.

Along the vertical orbit (figure 11) the deviation peaks near the first null, at −33° and +32° with the plain dome: 8.3 and 7.4 dB RMS with LR2, 7.6 and 7.0 with LR4, 6.5 and 6.1 with LR8, 2.5 and 2.7 with the brickwall. The floor bounces of the 300 rooms sit on the lower peak; the ceiling bounces on the upper one's shoulder.

6.3 Results

The averages flatter the low-slope crossover (table 5, figure 12). At the crossover frequency the four crossovers give the same response, each driver at half amplitude and in phase; they differ by the width of the band where both drivers play. In the rooms' averaged response the plain dome's LR2 deviates by 1.11 dB RMS, the LR4 by 1.36, the brickwall by 1.32; the deepest dips are 1.88, 2.37 and 3.01 dB, the brickwall's being its narrow notch at the crossover. With the plain dome the brickwall also turns the LR4's broad dip into a step of about 3 dB to the dome's wider radiation, and the averaged curve is no smoother. With the waveguide there is no step: 0.95 against 0.48 dB (0.81 with LR2). The PIR ranks the crossovers the same way.

The paths rank them the other way (table 6, figure 13). The floor and ceiling bounces carry 5.51 dB RMS with LR2, 4.58 with LR4 and 1.86 with the brickwall (plain dome; 4.54, 3.99 and 1.58 with the waveguide): from LR4 to the brickwall, 59 and 60 % less; LR2 is 20 % worse than LR4. For all the first reflections the brickwall removes 38 and 53 %, for every path in steady state 28 and 50 %. The deepest dip of the floor and ceiling bounces changes little (12.7 against 10.4 dB with the plain dome, 10.7 against 11.7 dB with the waveguide): the brickwall's notches are as deep, ten to thirteen times narrower, about a thirtieth of an octave against a third.

How much the averages hide. With the plain dome and LR4, the floor and ceiling bounces carry 4.58 dB RMS within an octave of the crossover and the rooms' averaged response shows 1.36 dB: the averaging divides the colouration by about 3.4. With LR2 it divides it by 5 (5.51 against 1.11 dB), with the brickwall by 1.4. The lower the slope, the wider the band where the drivers interfere, the more directions see the notch at different frequencies, and the more the average fills it.

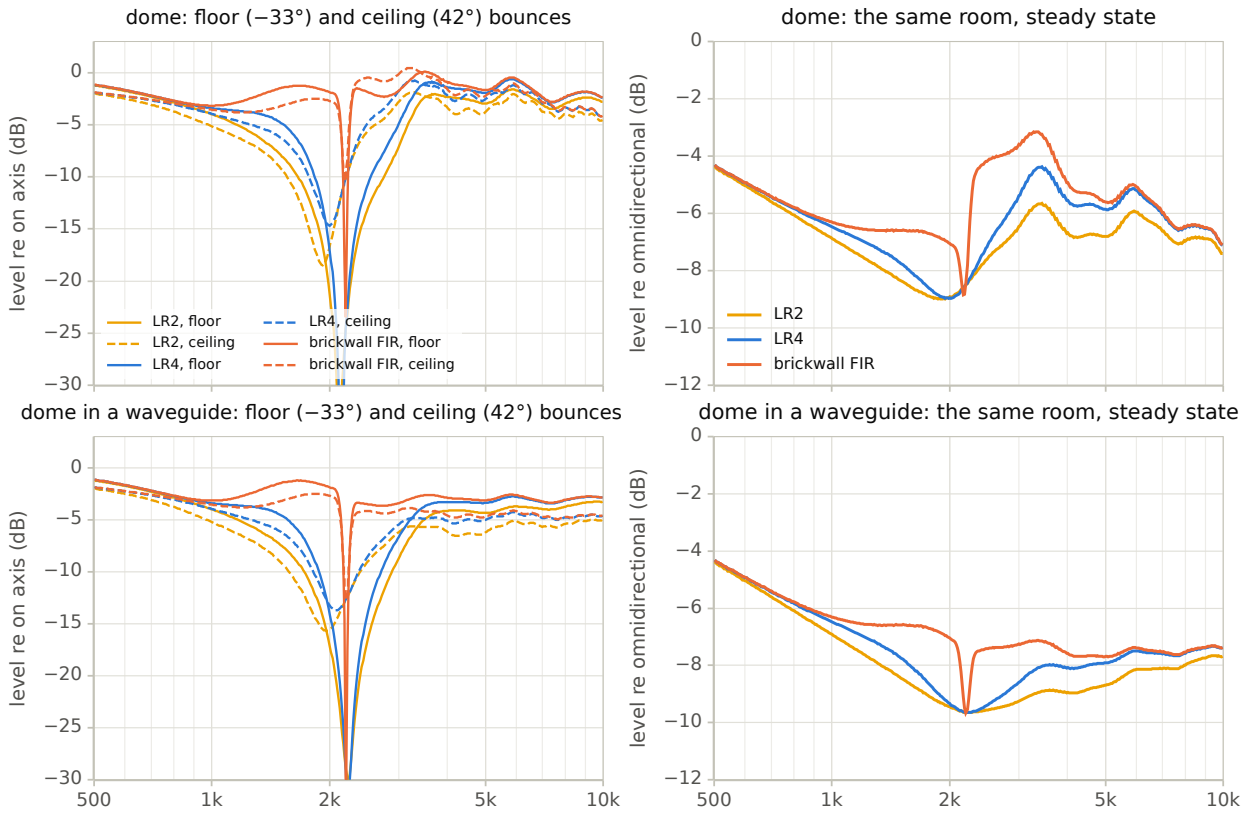


Figure 10: The living room of figure 2. Left: the loudspeaker's response in the directions of the floor and ceiling bounces, relative to the axis, with three crossovers. Right: the same room's steady-state response, relative to an omnidirectional source.

The standard's weights give the paths' colouration less weight than the rooms do (1.73 against 2.25 dB with the plain dome and LR4), because the PIR leans on the Listening Window, where vertical lobing does not show; they show a slightly smaller gain from the brickwall (22 against 28 % with the plain dome, 47 against 50 % with the waveguide). The updated weights sit in between.

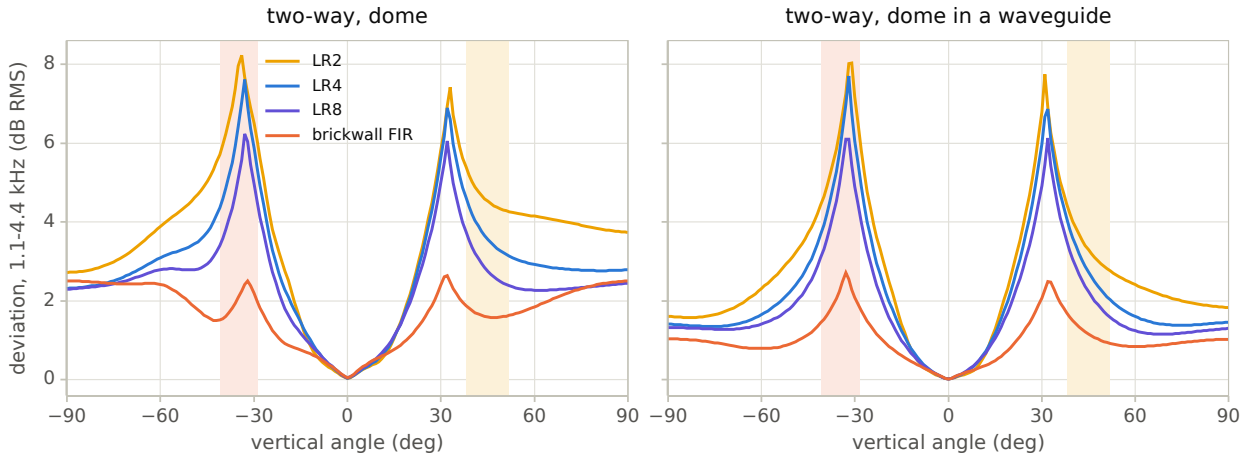


Figure 11: Deviation of the response in each direction of the vertical orbit, within one octave of the crossover, and the directions of the rooms' floor and ceiling bounces (shaded, 10–90 %).

Table 5: Averaged curves within one octave of the crossover (dB): RMS deviation from the straight line of the PIR, of the PIR with the fitted weights, of the rooms' steady state and early sound (median of the 300); deepest dip of the PIR and of the rooms' steady state.

	RMS deviation				deepest dip	
	PIR	fitted	rooms steady	rooms early	PIR	rooms steady
dome, upright, LR2	0.77	0.97	1.11	0.59	1.21	1.88
dome, upright, LR4	0.96	1.24	1.36	0.70	1.56	2.37
dome, upright, LR8	1.09	1.39	1.48	0.78	1.81	2.72
dome, upright, brickwall FIR	1.04	1.29	1.32	0.69	2.02	3.01
dome, on its side, LR2	0.74	1.12	1.10	0.51	1.21	1.88
dome, on its side, LR4	0.99	1.40	1.37	0.66	1.65	2.43
dome, on its side, LR8	1.15	1.51	1.51	0.82	1.95	2.80
dome, on its side, brickwall FIR	1.09	1.24	1.34	0.85	2.22	3.12
waveguide, upright, LR2	0.51	0.68	0.81	0.37	0.84	1.35
waveguide, upright, LR4	0.64	0.85	0.95	0.45	1.24	1.84
waveguide, upright, LR8	0.63	0.83	0.92	0.44	1.51	2.18
waveguide, upright, brickwall FIR	0.35	0.43	0.48	0.24	1.75	2.49
waveguide, on its side, LR2	0.59	1.00	0.85	0.26	0.93	1.36
waveguide, on its side, LR4	0.75	1.17	1.00	0.41	1.42	1.91
waveguide, on its side, LR8	0.73	1.12	0.98	0.44	1.74	2.28
waveguide, on its side, brickwall FIR	0.38	0.55	0.50	0.28	2.02	2.62

Table 6: Colouration of the paths, upright two-way: RMS deviation within one octave of the crossover (dB), weighted by the PIR, by the fitted weights, and by the rooms (median of the 300): every path, the first reflections, the floor and ceiling bounces; and the deepest dip of the floor and ceiling bounces.

	PIR weights	fitted weights	rooms steady	rooms reflections	rooms floor, ceiling	floor, ceiling deepest dip
dome, LR2	1.87	2.20	2.41	3.13	5.51	15.2
dome, LR4	1.73	2.11	2.25	2.75	4.58	12.7
dome, LR8	1.65	2.02	2.13	2.44	3.67	11.7
dome, brickwall FIR	1.35	1.58	1.62	1.70	1.86	10.4
waveguide, LR2	1.37	1.63	1.80	2.37	4.54	11.3
waveguide, LR4	1.19	1.50	1.65	2.11	3.99	10.7
waveguide, LR8	1.04	1.34	1.45	1.81	3.35	10.9
waveguide, brickwall FIR	0.64	0.76	0.83	0.99	1.58	11.7

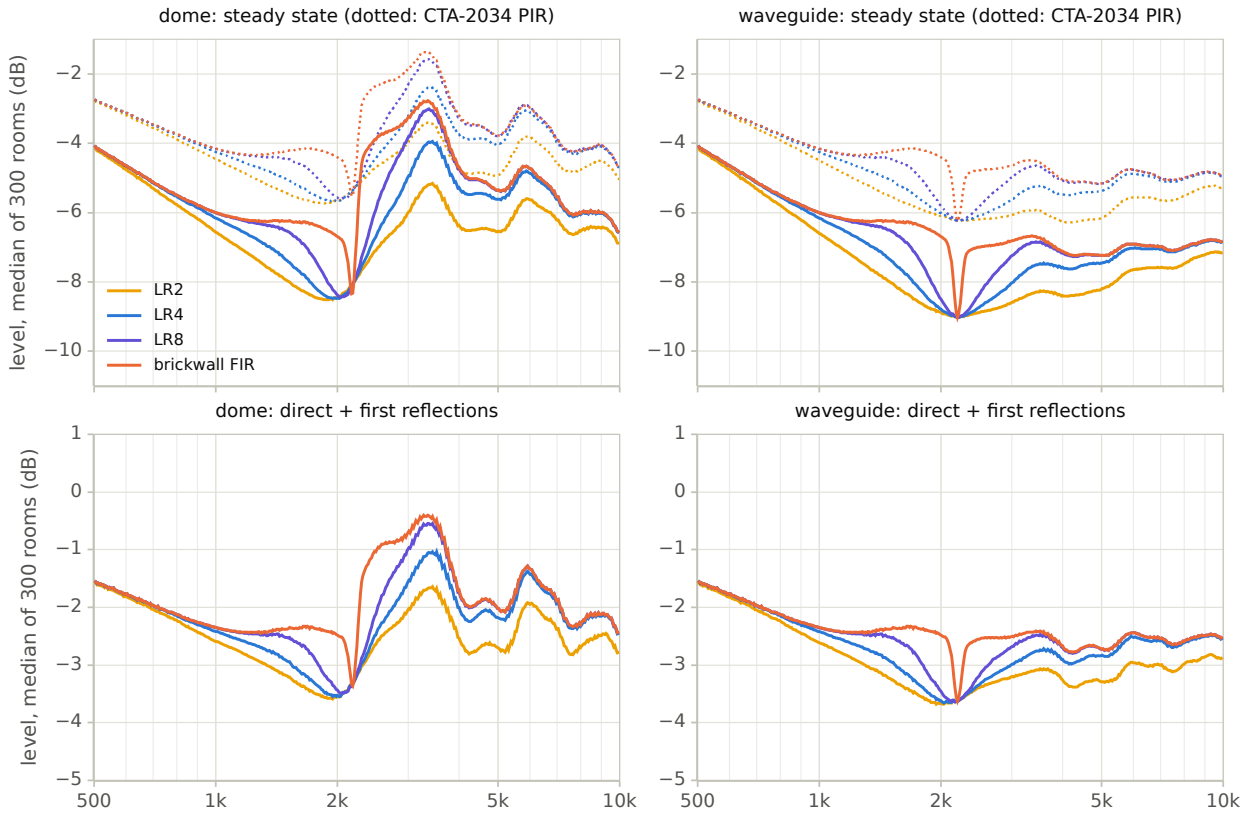


Figure 12: Median in-room response of the 300 rooms with the four crossovers, steady state (top, the PIR dotted) and early sound (bottom); plain dome (left), dome in a waveguide (right).

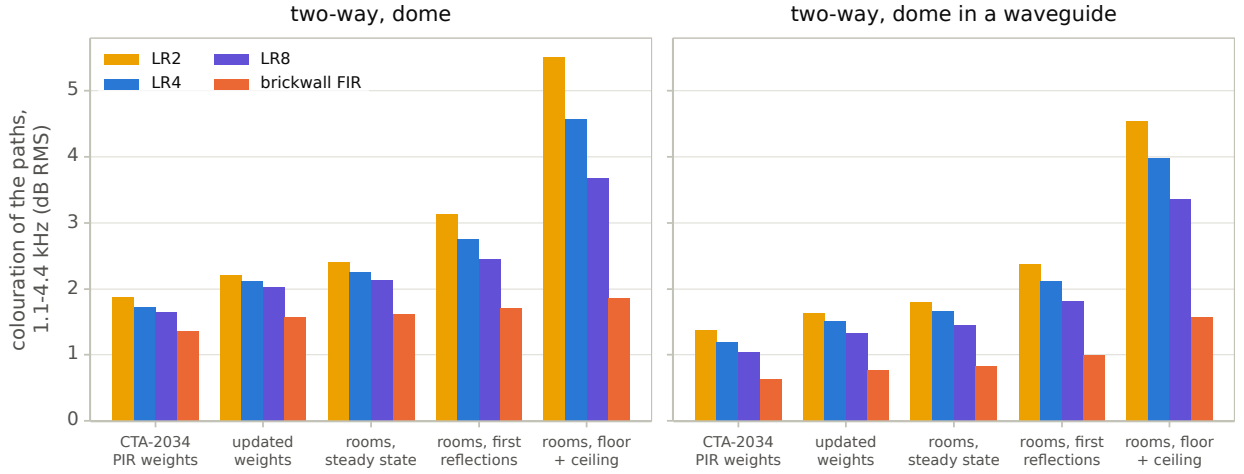


Figure 13: Colouration of the paths with the four crossovers, upright two-way.

Table 7: Deviation within one octave of the crossover (dB RMS, median of the 300 rooms) of the paths added in power at the reference point, and with their phases: at one point (median of the fourteen), at the two ears averaged in power, at the fourteen points averaged. Upright two-way; responses smoothed by the auditory filters.

	steady state				early sound			
	power	point	ears	14 pts	power	point	ears	14 pts
dome, LR2	1.08	1.12	1.10	1.07	0.56	0.89	0.81	0.73
dome, LR4	1.32	1.36	1.33	1.31	0.67	1.03	0.96	0.86
dome, LR8	1.41	1.46	1.44	1.41	0.73	1.16	1.09	0.97
dome, brickwall FIR	1.19	1.27	1.25	1.22	0.61	1.23	1.15	0.98
waveguide, LR2	0.80	0.87	0.85	0.82	0.36	0.67	0.60	0.52
waveguide, LR4	0.93	1.01	0.99	0.96	0.44	0.79	0.72	0.63
waveguide, LR8	0.88	0.98	0.95	0.91	0.41	0.87	0.80	0.69
waveguide, brickwall FIR	0.37	0.58	0.54	0.47	0.17	0.91	0.83	0.67
omnidirectional source	0.00	0.39	0.31	0.20	0.00	1.32	1.12	0.78

6.4 Interference at the ears

Every response so far adds the paths in power, as if each were heard on its own. At a fixed point the direct sound and the reflections add as pressures: a reflection delayed by τ digs a notch every $1/\tau$ into the response, a comb. To see what the comb does to the comparison, the direct sound and the images up to the third order were added with their phases at fourteen points: the two ears, 17.5 cm apart and without a head, of seven head positions (the reference one, 10 cm to either side, 10 cm forward and back, 5 cm up and down). Each image keeps the square root of its energy weight as a real, positive reflection factor, and the loudspeaker's response in the direction its path leaves in; the diffuse field is still added in energy. The ear does not resolve the fine comb of a late reflection, so every response is smoothed by the auditory filters of Glasberg and Moore [17], rounded exponentials one ERB wide (262 Hz at the crossover, about a sixth of an octave), computed on a 4 Hz grid, before its deviation within one octave of the crossover is taken. The images' energy matches the study's within 10^{-6} dB. An omnidirectional source in the same rooms shows the combs the rooms produce on their own.

In steady state the phases change little (table 7). The diffuse field, added in energy, fills the combs: at one point the deviation grows by 0.04 to 0.21 dB, and the crossovers keep their order (plain dome: LR2 1.12, LR4 1.36, brickwall 1.27 dB; waveguide: 1.01 with LR4 against 0.58 with the brickwall). The omnidirectional source's combs leave 0.39 dB.

The early sound at one point is another matter (figures 14 and 15). There the direct sound meets the floor and ceiling bounces, 1.9 and 3.6 ms later in the median room: their combs, a few hundred hertz apart, are about as wide as the auditory filters and survive the smoothing. The omnidirectional source shows 1.32 dB RMS within an octave of the crossover at one point, more than any crossover in power (0.73 dB at most, LR8 with the plain dome). The loudspeakers fall in between, from 0.89 (LR2) to 1.23 dB (brickwall) with the plain dome, from 0.67 to 0.91 dB with the waveguide, and the brickwall comes last: the smoothest in power with the waveguide (0.17 against 0.44 dB for LR4), it is the roughest at one point. The Linkwitz-Riley notch takes the floor and ceiling bounces out around the crossover, and their comb with them: in the living room the fourteen points close in on the energy sum around 2 kHz with LR4, not with the brickwall, which leaves the reflections and their comb intact. Averaging the two ears or the fourteen points narrows the gaps without closing them (0.63 against 0.67 dB over the fourteen points with the waveguide).

What the comb measures is the room, not the crossover: a source without a crossover shows it everywhere, and a crossover that notches the reflections near its frequency removes it there, with part of the reflections' energy. Whether that sounds better is for listening tests to tell. A single reflection's comb colours the sound less for two ears than for one (binaural decoloration [18]), and

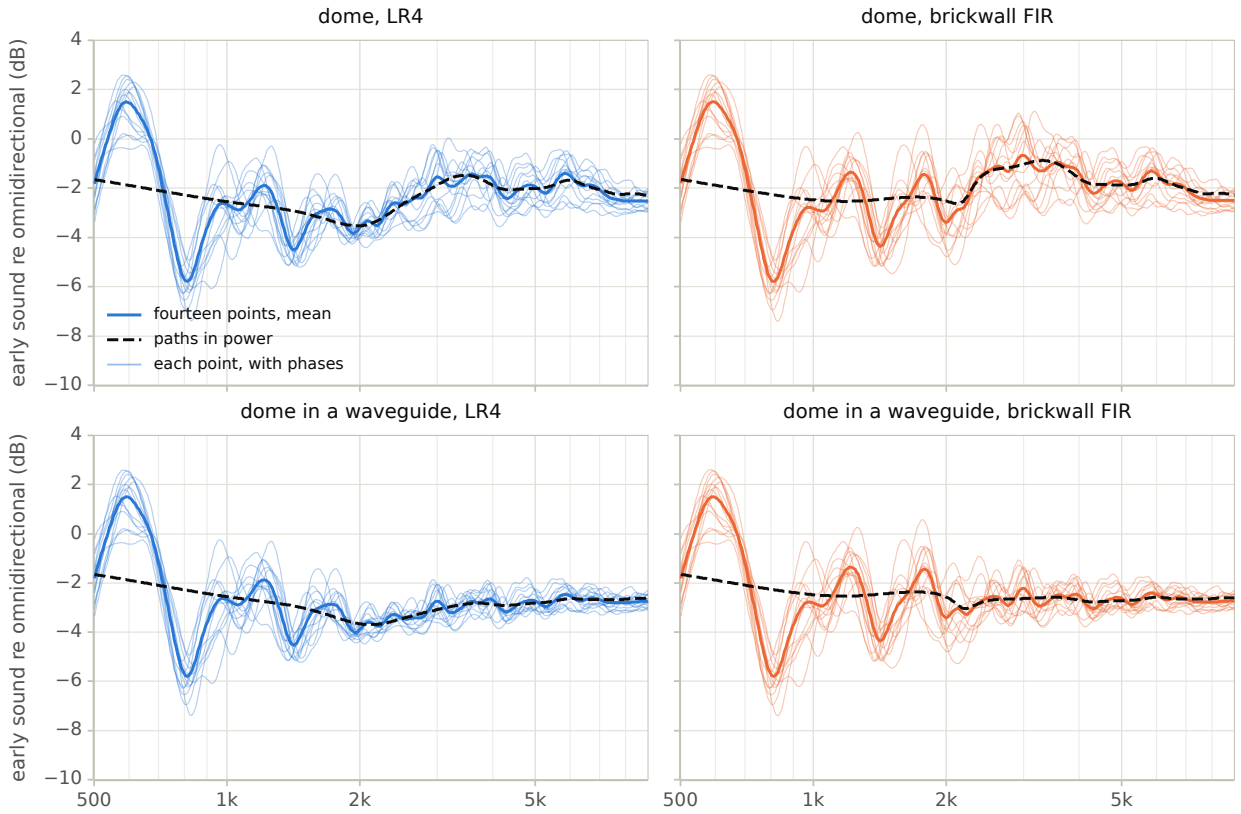


Figure 14: Early sound in the living room of figure 2, relative to an omnidirectional source: the paths added in power at the reference point (dashed), and with their phases at each of the fourteen points (thin) and averaged over them (thick); smoothed by the auditory filters.

the reflections' own spectrum, which the colouration of the paths measures, is what the ear ascribes to the room. Interference thus leaves the steady-state picture as it is; at a fixed point it trades the crossover's colouration of the reflections for the comb those reflections cause.

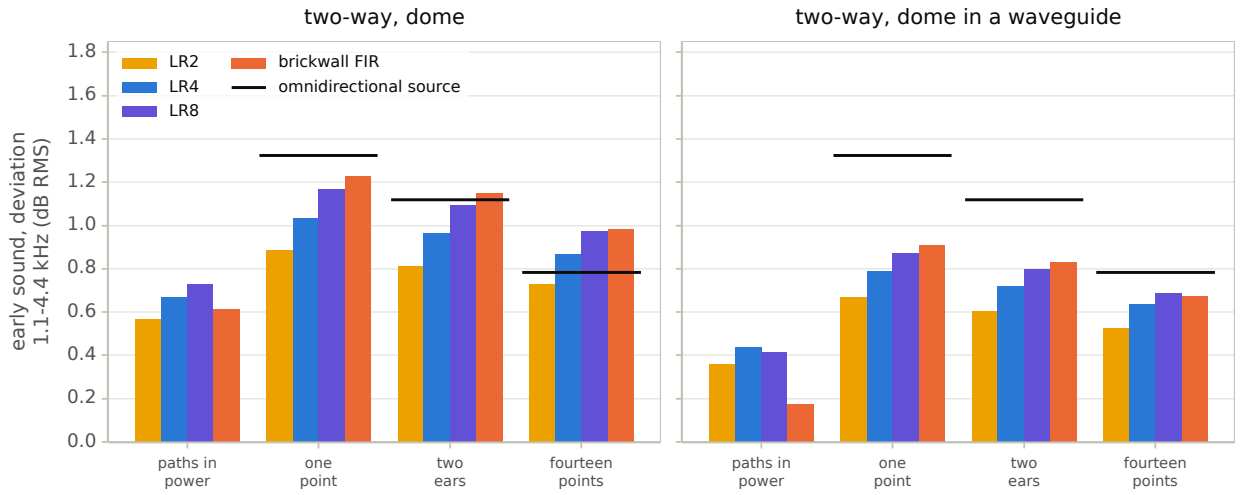


Figure 15: Deviation of the early sound within one octave of the crossover (median of the 300 rooms): the paths added in power, then with their phases at one point, at the two ears averaged, at the fourteen points averaged. Black lines: an omnidirectional source, the combs of the rooms alone.

7 Discussion

7.1 The three objections

The vertical weight. In these rooms the floor and ceiling bring as large a share of the first reflections' energy as the Early Reflections give them, and their bounces leave where the standard looks. They do arrive first, and where the side walls are treated they bring half of it, a little more than the standard's 40 %. The objection holds for the sound power: the two orbits weigh the plane without interference as much as the one with it, and miss half of the power lost to lobing. That bias is the same for horizontal and vertical lobing; for an upright loudspeaker it falls on the vertical one.

The Listening Window. Its narrow vertical extent is not what hides the lobing: the direct sound of a seated listener does not see vertical lobing either, and laying the loudspeaker on its side shows what the window guards against. What hides it is the averaging every curve shares, the room's own average included. The floor and ceiling bounces of an LR4 two-way carry notches of 10 to 40 dB, a third of an octave wide, which fall at different frequencies in different directions; the averages turn them into a dip of 2 dB. The lower the slope, the more the averages flatter the crossover: the in-room curve and the PIR rank LR2 as the smoothest of the Linkwitz-Riley crossovers, while its reflections are the most coloured. In that form, the objection holds.

The brickwall crossover. It reduces the colouration of the floor and ceiling bounces by about 60 %, with or without the waveguide, and of all the first reflections by 38 to 53 %. Its notches are as deep as the Linkwitz-Riley ones, ten to thirteen times narrower. Whether the in-room curve becomes smoother depends on the drivers: where the woofer beams at the crossover and the tweeter does not, the brickwall turns the lobing dip into a step of directivity. Matching the directivities at the crossover (a waveguide) and narrowing the band where both drivers play (a steep crossover) address two different causes of the off-axis colouration.

7.2 An updated weighting

What the rooms ask for is simple to state:

- the sound power over the sphere, which near-field scanners and spherical measurements now give, instead of two orbits;
- for the steady-state in-room response, about $0.1\,\mathrm{LW} + 0.9\,\mathrm{SP}$: the room behaves mostly as a sound-power meter, the direct sound adding a little forward weight;

- for the early sound, which carries the reflections heard with the direct sound, about $0.4\,\mathrm{LW} + 0.5\,\mathrm{ER} + 0.1\,\mathrm{SP}$.

Fitted to the received energy, the first set of weights predicts the rooms within their own spread. It does not undo the averaging, though: no weighted mean of power responses shows the notch a single reflection carries. That has to be read direction by direction, on the polar maps or with a measure like the colouration of the paths used here, in the directions where the floor and ceiling bounces leave, -30 to $-40°$ and $+40$ to $+50°$ at usual listening distances.

7.3 Limits

The rooms are shoeboxes, with uniform absorption per surface except for the patches, textbook coefficients and a hybrid specular-diffuse model; real rooms have furniture in the paths, openings, irregular shapes. Their reverberation and their energy shares follow from the room constant and are not specific to this model, but the share of each first reflection depends on the materials at the reflection point. Paths are added in energy, except in section 6.4, where the images meet with their phases at points without a head; the room's modes below about 200 Hz are outside the study. The early sound is an energy sum too; hearing weighs reflections differently from their energy (binaural decoloration of lateral reflections, the salience of the floor bounce in the median plane [14, 15]), which no weighting of power responses captures. The colouration metrics are physical measures: a narrow notch is harder to hear than a broad one of the same depth [16], and the RMS deviation accounts for that only through its width. The loudspeaker is one two-way bookshelf in two versions; a three-way, a cardioid, a floor-standing loudspeaker or a larger listening distance would change the numbers, not the method. Finally the study uses CTA-2034-A; its revision B was not checked.

8 Conclusion

Seen as weights over the directions of the sphere, CTA-2034's curves and a room can be compared directly. In 300 simulated domestic rooms the standard looks for the first reflections in the right directions and gives the floor and ceiling the right share of them; its Listening Window matches where listeners sit. Its weights do not match the energy received: the first reflections carry 11 % of the steady-state energy, not 44 %, and the rooms are predicted to within their own spread by about $0.1\,\mathrm{LW} + 0.9\,\mathrm{SP}$ with the sound power of the whole sphere, which two orbits underestimate where drivers interfere. Any average, the room's included, fills the notches the crossover digs into each reflection, and flatters the low-slope crossovers most. Read path by path, a brickwall crossover removes about 60 % of the colouration of the floor and ceiling bounces; it gives a smoother in-room response only when the drivers' directivities match at the crossover. Added with their phases at the ears, the paths leave the steady state as it is; at a fixed point the early sound carries the comb of the floor and ceiling bounces, which a Linkwitz-Riley notch removes near the crossover along with the reflections, and which hearing with two ears attenuates.

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